

NAME: _____

TEACHER: _____

THE SCOTS COLLEGE



YEAR 12

TRIAL HSC EXAMINATION 2007

MATHEMATICS

EXTENSION 1

TIME ALLOWED: TWO HOURS
[plus 5 minutes reading time]

INSTRUCTIONS:

- Attempt **ALL** questions.
- **ALL** questions are of equal value.
- **ALL** necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are attached.
- Board approved calculators **ONLY** may be used.
- Answer each question in a **SEPARATE** Writing Booklet.
- Additional Writing Booklets are available if you require them.

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

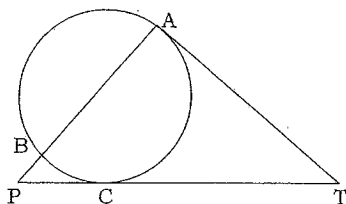
QUESTION 1 [12 MARKS]

- (a) The interval CD has end points C(-2, 3) and D(10, 11). Find the co-ordinates of the point P which divides the interval CD internally in the ratio of 3:1. [2]
- (b) Solve the inequality $\frac{1}{x^2-1} < 0$ algebraically and show your solution on a number line. [2]
- (c) The graphs $y=x$ and $y=x^3$ intersect at $x=1$. Find the size of the acute angle between these curves at this point of intersection (to the nearest degree). [3]
- (d) Find the exact value of $\int_0^{\sqrt{5}} \frac{1}{\sqrt{5-x^2}} dx$ [2]
- (e) By using the value of $x=0.5$ as a first approximation, find a root of the equation $x+\ln x=0$ using Newton's method to find a second approximation. Give your answer correct to 2 decimal places. [3]

QUESTION 2 [12 MARKS] START A NEW WRITING BOOKLET

- (a) Prove the identity $\frac{\sin A}{\sin A + \cos A} - \frac{\sin A}{\sin A - \cos A} = \tan 2A$. [3]

- (b) [5]



AB is the diameter of a circle ABC. The tangents at A and C meet at T. The lines TC and AB are produced to meet at P.

- (i) Copy the diagram into your examination booklet. Join AC and CB.
- (ii) Prove that $\angle CAT = 90^\circ - \angle BCP$
- (iii) Hence, or otherwise, prove that $\angle ATC = 2\angle BCP$
- (c) (i) State the domain and range of the function $y = \cos^{-1} \frac{x}{2}$
- (ii) Sketch the graph of the function given by $y = \cos^{-1} \frac{x}{2}$
- (iii) Find the equation of the tangent to the curve at the point where it cuts the y axis. [4]

QUESTION 3 [12 MARKS] START A NEW WRITING BOOKLET

- (a) Consider the circle with equation $x^2 + y^2 - 2x - 14y + 25 = 0$. [6]
- (i) Determine the co-ordinates of its centre and find the length of the radius.
- (ii) Show that if the line $y=kx$ intersects the circle at two distinct points, then:

$$(1+7k)^2 > 25(1+k^2)$$
- (iii) Find all values of k for which the line $y=kx$ is a tangent to the circle.
- (b) Evaluate $\int_0^{\frac{\pi}{2}} \sin^2 3x dx$. [2]
- (c) It is known that the polynomial $p(x) = x^3 + ax^2 + bx + c$ has a relative minimum at $x=\beta$ and a relative maximum at $x=\alpha$. [4]
- (i) Prove $\alpha + \beta = -\frac{2}{3}a$
- (ii) Show that the point of inflexion occurs at $x = \frac{\alpha + \beta}{2}$

QUESTION 4 [12 MARKS] START A NEW WRITING BOOKLET

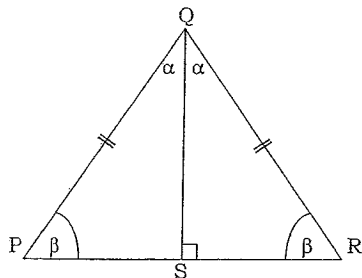
- (a) Using the substitution $u=2x+1$, or otherwise, find the exact value of $\int_0^1 \frac{4x}{2x+1} dx$ [3]
- (b) The function $f(x)$ is given by $f(x) = \sin^{-1} x + \cos^{-1} x$, $0 \leq x \leq 1$. [3]
- (i) Find $f'(x)$
- (ii) Sketch the graph $y=f(x)$
- (c) (i) By equating the coefficients of $\sin x$ and $\cos x$, or otherwise, find the values of the constants A and B which satisfy the identity

$$A(2\sin x + \cos x) + B(2\cos x - \sin x) \equiv \sin x + 8\cos x$$
- (ii) Using (i), find $\int \frac{\sin x + 8\cos x}{2\sin x + \cos x} dx$

QUESTION 5 [12 MARKS]

START A NEW WRITING BOOKLET

(a)



The triangle PQR is isosceles with $PQ = RQ$ and QS is perpendicular to PR .

Let $\angle PQS = \angle RQS = \alpha$ and $\angle QPS = \angle QRS = \beta$

(i) Show that $\cos \alpha = \sin \beta$.

(ii) Using the sine rule for triangle PQR, show that $\sin 2\alpha = 2 \sin \alpha \cos \alpha$.

(iii) Given that $0 < \alpha < \frac{\pi}{2}$, show that the limiting sum of the geometric series $\sin 2\alpha + \sin 2\alpha \cos^2 \alpha + \sin 2\alpha \cos^4 \alpha + \sin 2\alpha \cos^6 \alpha + \dots$ is equal to $2 \cot \alpha$.

(b) Find the value of the term that is independent of x in the expansion of $\left(x^2 - \frac{2}{x}\right)^6$.

(c) Find all angles θ , where $0 \leq \theta \leq 2\pi$, for which $\sqrt{3} \cos \theta - \sin \theta = 1$.

[5]

[3]

[4]

QUESTION 6 [12 MARKS]

START A NEW WRITING BOOKLET

(a) Consider the parabola $x^2 = 4ay$, where $a > 0$, and let the tangents at $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ intersect at the point T . Let the focus of the parabola be S , with co-ordinates $(0, a)$.

[6]

(i) Show that the equation of the tangent at P is $y = px - ap^2$.

(ii) Find the co-ordinates of T .

(iii) Show that $SP = a(p^2 + 1)$.

(iv) Suppose that P and Q move on the parabola $x^2 = 4ay$ in such a manner that $SP + SQ = 4a$.

Show that the locus of the point T is a parabola and write down the co-ordinates of the vertex and the focus of this parabola.

(b) A particle is moving in simple harmonic motion about a fixed point O . Its period is 4π seconds and its amplitude is 3cm.

Find its speed at the point O .

[2]

(c) Let T be the temperature inside a building at time t and let T_0 be the constant air temperature outside the building. Newton's law of cooling states that the rate of change of the temperature t is proportional to $T - T_0$.

(i) Show that $T = T_0 + Ce^{kt}$ (where C and k are constants) satisfies Newton's law of cooling.

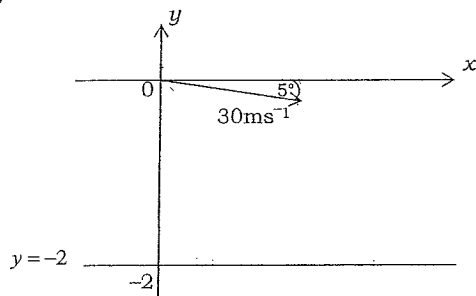
(ii) The outside air temperature is 5°C and a break down in the heating system causes the temperature inside the building to drop from 20°C to 17°C in half an hour. After how many hours, correct to 2 decimal places, is the temperature inside the building equal to 10°C ?

QUESTION 7 [12 MARKS]

START A NEW WRITING BOOKLET

- (a) A particle moves along the x axis. Its velocity v at position x is given by $v = \sqrt{10x - x^2}$. Find the acceleration of the particle when $x = 2$. [2]

- (b) [6]



A tennis ball leaves the player's racquet 2 metres above the ground with a velocity of 30ms^{-1} at an angle of 5° below the horizontal. The equations of motion for the ball are $\ddot{x} = 0$ and $\ddot{y} = -10$.

Take the origin to be the point where the ball leaves the player's racquet.

- (i) Using calculus, show that the co-ordinates of the ball at time t are given by:
 $x = 30t \cos 5^\circ$
 $y = -30t \sin 5^\circ - 5t^2$
- (ii) Find the time at which the ball strikes the ground, correct to 2 decimal places.
- (iii) Calculate the angle, to the nearest degree, at which the ball strikes the ground.

- (c) (i) For positive integers n and r , with $n > r$, show that [4]

$${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$$

where ${}^nC_r = \frac{n!}{r!(n-r)!}$. Do NOT use induction.

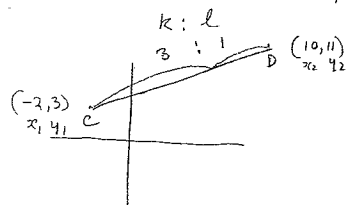
- (ii) Use mathematical induction to prove that, for $n \geq 3$,

$$\sum_{j=3}^n {}^{j-1}C_2 = {}^nC_3$$

END OF PAPER

Q1

(a)



HE1

$$\begin{aligned} x_p &= \frac{kx_2 + lx_1}{k+l} \\ &= \frac{3(10) + 1(-2)}{3+1} \\ &= \frac{28}{4} \\ &= 7 \end{aligned}$$

$$\begin{aligned} y_p &= \frac{kx_2 + lx_1}{k+l} \\ &= \frac{3(11) + 1(3)}{3+1} \\ &= \frac{36}{4} \\ &= 9. \end{aligned}$$

(1, 1)

(2)

∴ P has co-ordinates (7, 9)

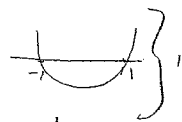
(b) $\frac{1}{x^2-1} < 0 \rightarrow \frac{1}{(x+1)(x-1)} < 0$

$$\frac{(x+1)^2(x-1)^2}{(x+1)(x-1)} < 0 \quad [(x+1)^2(x-1)^2]$$

$$(x+1)(x-1) < 0$$

$$\therefore \{-1 < x < 1\}$$

Graphical solution :



(3)

(c) $l_1: y = x$
gradient $m_1 = 1$

$l_2: y = x^5$
 $y' = 5x^4$
 $\therefore m_2 = 5$ at $x = 1$.

$$\begin{aligned} \text{acute angle } \tan \theta &= \frac{m_2 - m_1}{1 + m_1 m_2} \\ &= \frac{5 - 1}{1 + 5(1)} \\ &= \frac{4}{6} \\ &= \frac{2}{3} \end{aligned}$$

$$\begin{aligned} \theta &= \tan^{-1}\left(\frac{2}{3}\right) \\ &= 33.69 \\ &= 34^\circ \text{ (nearest minute)} \end{aligned}$$

(2)

(2)

$$\begin{aligned} &\int_0^{\sqrt{5}} \frac{1}{\sqrt{(\sqrt{5})^2 - x^2}} \\ &= \left[\sin^{-1} \frac{x}{\sqrt{5}} \right]_0^{\sqrt{5}} \\ &= \sin^{-1} \frac{\sqrt{5}}{\sqrt{5}} - \sin^{-1} \frac{0}{\sqrt{5}} \\ &= \sin^{-1} 1 - \sin^{-1} 0 \\ &= \frac{\pi}{2} \end{aligned}$$

HE4

(e)

let $f(x) = x + \ln x$

$$f'(x) = 1 + \frac{1}{x}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

using $x_1 = 0.5$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.5 - \frac{0.5 + \ln 0.5}{1 + \frac{1}{0.5}}$$

$$= 0.5 - \frac{0.5 + \ln(0.5)}{3}$$

$$= 0.5643$$

$$= 0.56 \text{ (2dp)}$$

PE1

(2)

Q2

$$(a) \frac{\sin A}{\sin A + \cos A} - \frac{\sin A}{\sin A - \cos A}$$

$$= \sin A \left(\frac{(\sin A - \cos A) - (\sin A + \cos A)}{\sin^2 A - \cos^2 A} \right)$$

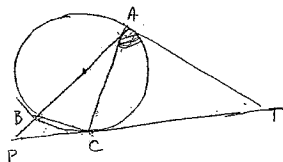
$$= \frac{\sin A (-2 \cos A)}{\sin^2 A - \cos^2 A}$$

$$\text{PE 2} = \frac{2 \sin A \cos A}{\cos^2 A - \sin^2 A}$$

$$= \frac{\sin 2A}{\cos 2A}$$

$$= \tan 2A$$

(b) (i)

(ii) $\angle BCP = \angle BAC$ (alt. segment theorem)PE 3 $\angle BAT = 90^\circ$ (angle between radius/tangent)

$$\angle BAT = \angle BAC + \angle CAT = \angle BCP + \angle CAT$$

$$\angle CAT = \angle BAT - \angle BCP$$

$$\angle CAT = 90^\circ - \angle BCP$$

(iii) $\angle ACT = \angle CAT$ (base angles of triangle from an ext. pt)

$$\angle ATC + \angle ACT + \angle CAT = 180^\circ \text{ (angle sum of } \triangle)$$

$$\angle ATC + (90^\circ - \angle BCP) + (90^\circ - \angle BCP) = 180^\circ$$

$$\angle ATC - 2\angle BCP = 0$$

$$\angle ATC = 2\angle BCP$$

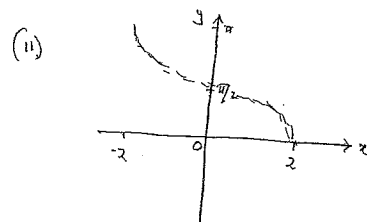
(3)

$$(c) (i) y = \cos^{-1} \frac{x}{2}$$

$$D: \{-1 \leq \frac{x}{2} \leq 1\}$$

$$D = \{-2 \leq x \leq 2\}$$

$$R \neq \{0 \leq y \leq \pi\} \quad (2)$$

(iii) The curve cuts the y-axis at $x=0$.

$$y = \cos^{-1} \frac{x}{2}$$

HE 5

$$y' = - \frac{1}{\sqrt{1 - (\frac{x}{2})^2}} \times \frac{1}{2}$$

$$= - \frac{1}{2\sqrt{1 - 0}}$$

$$= -\frac{1}{2}$$

$$\text{eq of tangent } y - y_1 = m(x - x_1)$$

$$\text{eq cuts y axis at } (0, \frac{\pi}{2})$$

$$\therefore \text{eq: } y - \frac{\pi}{2} = -\frac{1}{2}(x - 0)$$

$$y = -\frac{1}{2}x + \frac{\pi}{2}$$

(4)

Q3.

$$\begin{aligned}
 (a) \quad (1) \quad x^2 - 2x + y^2 - 14y &= -25 \\
 x^2 - 2x + 1 + y^2 - 14y + 49 &= -25 + 1 + 49 \\
 (x-1)^2 + (y-7)^2 &= 25 \\
 \therefore \text{centre } (1, 7) \text{ and radius} &= 5
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \text{Solving } y = kx \text{ and } x^2 - 2x + y^2 - 14y + 25 &= 0 \\
 x^2 - 2x + (kx)^2 - 14kx + 25 &= 0 \\
 x^2 + k^2x^2 - x(2+14k) + 25 &= 0 \\
 (1+k^2)x^2 - (2+14k)x + 25 &= 0
 \end{aligned}$$

Two distinct real roots if $\Delta > 0$

$$\begin{aligned}
 \Delta &= [2(1+7k)]^2 - 4(1+k^2)25 > 0 \\
 4(1+7k)^2 - 4 \cdot 25(1+k^2) &> 0 \\
 (1+7k)^2 - 25(1+k^2) &> 0 \\
 (1+7k)^2 &> 25(1+k^2)
 \end{aligned}$$

(iii) The line $y = kx$ is tangential if $(1+7k)^2 = 25(1+k^2)$, $\Delta = 0$

$$\begin{aligned}
 (1+7k)^2 &= 25 + 25k^2 \\
 1 + 14k + 49k^2 &= 25 + 25k^2 \\
 24k^2 + 14k - 24 &= 0 \\
 12k^2 + 7k - 12 &= 0 \\
 (4k-3)(3k+4) &= 0 \\
 \therefore k &= \frac{3}{4}, -\frac{4}{3}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \int_0^{\pi/2} \sin^2 3x \, dx \\
 = \int_0^{\pi/2} \frac{1}{2}(1 - \cos 6x) \, dx \\
 = \frac{1}{2} \left[x - \frac{\sin 6x}{6} \right]_0^{\pi/2} \\
 = \frac{1}{2} \left[\left(\frac{\pi}{2} - \frac{\sin 3\pi}{6} \right) - 0 \right] \\
 = \frac{1}{2} \left[\left(\frac{\pi}{2} - 0 \right) - 0 \right] \\
 = \frac{1}{2} \left[\frac{\pi}{2} \right] \\
 = \frac{\pi}{4}
 \end{aligned}$$

(5)

$$Q3(c) (i) \quad p(x) = x^3 + ax^2 + bx + c.$$

$$p'(x) = 3x^2 + 2ax + b$$

turning points at $x = \alpha, \beta$ when $p'(x) = 0$

$$p'(\alpha) = 3\alpha^2 + 2a\alpha + b = 0$$

$$p'(\beta) = 3\beta^2 + 2a\beta + b = 0$$

$$\therefore 3\alpha^2 + 2a\alpha + b = 3\beta^2 + 2a\beta + b$$

$$3\alpha^2 - 3\beta^2 = 2a\beta - 2a\alpha$$

$$3(\alpha - \beta)(\alpha + \beta) = 2a(\beta - \alpha)$$

$$3(\alpha + \beta) = -2a$$

$$\alpha + \beta = -\frac{2}{3}a$$

HE1

(ii) Inflection occurs when $p''(x) = 0$

$$p'' = 6x + 2a = 0$$

$$\therefore 2a = -6x$$

$$a = -3x$$

$$\therefore \alpha + \beta = -\frac{2}{3}a = -\frac{2}{3}(-3x) = 2x \quad \text{using (i)}$$

$$\therefore 2x = \alpha + \beta$$

$$x = \frac{\alpha + \beta}{2}$$

(6)

(4)

(6)

(2)

Q4

$$(a) \quad \begin{aligned} u &= 2x+1 \\ du &= 2dx \\ \text{at } x=0, u &= 1 \\ \text{at } x=1, u &= 3 \end{aligned}$$

$$\rightarrow \begin{aligned} 2x &= u-1 \\ 4x &= 2(u-1) \\ \text{and } dx &= \frac{1}{2} du \end{aligned}$$

Alternative

$$\int_0^1 \frac{4x}{2x+1} dx$$

$$= \int_0^1 \frac{4x+2}{2x+1} - \frac{2}{2x+1} dx$$

$$= \int_0^1 2 - \frac{2}{2x+1} dx$$

$$= \int_0^1 2 - \frac{d(2x+1)}{2x+1} dx$$

$$= [2x - \ln(2x+1)]_0^1$$

$$= (2 - \ln 3) - 0$$

$$= 2 - \ln 3$$

(3)

HE 8

$$\int_0^1 \frac{4x dx}{2x+1}$$

$$= \int_1^3 \frac{2(u-1)}{u} \cdot \frac{1}{2} du$$

$$= \int_1^3 \left(\frac{u-1}{u} \right) du$$

$$= \int_1^3 \left(1 - \frac{1}{u} \right) du$$

$$= (u - \ln u) \Big|_1^3$$

$$= (3 - \ln 3) - (1 - \ln 1)$$

$$= 2 - \ln 3 + 0$$

$$= 2 - \ln 3$$

$$(b) (i) f(x) = \sin^{-1} x + \cos^{-1} x$$

$$f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0$$

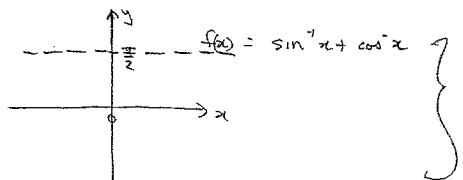
$$(ii) \text{ Since } f'(x) = 0, f(x) \text{ is a constant}$$

$$\text{at } x=0, f(0) = \sin^{-1} 0 + \cos^{-1} 0 = \frac{\pi}{2} + 0$$

$$f(x) = \frac{\pi}{2}$$

(3)

HE 1



4(e)

$$(i) \quad \left. \begin{aligned} A(2\sin x + \cos x) + B(2\cos x - \sin x) &\equiv \sin x + 8\cos x \\ 2A\sin x + A\cos x + 2B\cos x - B\sin x & \\ = (2A-B)\sin x + (A+2B)\cos x \end{aligned} \right\}$$

$$\therefore \begin{aligned} 2A-B &= 1 \\ A+2B &= 8 \end{aligned} \rightarrow \begin{aligned} 4A-2B &= 2 \\ A+2B &= 8 \\ \hline 5A &= 10 \\ A &= 2 \end{aligned}$$

$$2A-B=1$$

$$2(2)-B=1$$

$$4-1=B$$

$$\therefore B=3 \text{ and } A=2$$

(3)

Alternative

$$A(2\sin x + \cos x) + B(2\cos x - \sin x) \equiv \sin x + 8\cos x$$

$$\text{let } x=0, A(0+1) + B(2-0) = 8 \rightarrow A+2B=8$$

$$\text{let } x=\pi/2, A(2+0) + B(0-1) = 1 \rightarrow 2A-B=1$$

$$\therefore A=2, B=3 \text{ (see above)}$$

$$(ii) \sin x + 8\cos x = 2(2\sin x + \cos x) + 3(2\cos x - \sin x) \text{ from (i)}$$

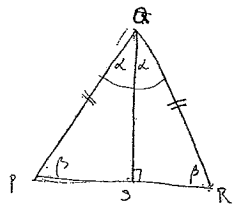
$$\therefore \frac{\sin x + 8\cos x}{2\sin x + \cos x} = \frac{2(2\sin x + \cos x)}{2\sin x + \cos x} + \frac{3(2\cos x - \sin x)}{2\sin x + \cos x}$$

$$\begin{aligned} \therefore \int \frac{\sin x + 8\cos x}{2\sin x + \cos x} dx &= \int \left[2 + 3 \left(\frac{2\cos x - \sin x}{2\sin x + \cos x} \right) \right] dx \\ &= \int \left[2 + 3 \frac{d(2\sin x + \cos x)}{2\sin x + \cos x} \right] dx \\ &= 2x + 3 \ln(2\sin x + \cos x) + C \end{aligned}$$

HE 1

(3)

Q 5.
(a)



$$(i) \frac{QS}{QP} = \sin \beta = \cos \alpha$$

$$(ii) \frac{PR}{\sin 2\alpha} = \frac{QP}{\sin \beta}$$

$$\frac{2PS}{\sin 2\alpha} = \frac{QP}{\sin \beta}$$

$$\frac{2PS}{QP} = \frac{\sin 2\alpha}{\sin \beta}$$

$$2 \cdot \sin \alpha = \frac{\sin 2\alpha}{\sin \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \sin \beta$$

$$\therefore \sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$(iii) \sin 2\alpha + \sin 2\alpha \cos^2 \alpha + \sin 2\alpha \cos^4 \alpha + \dots$$

$$= \sin 2\alpha (1 + \cos^2 \alpha + \cos^4 \alpha + \dots)$$

$$= \sin 2\alpha \left(\frac{1}{1 - \cos^2 \alpha} \right)$$

$$= \sin 2\alpha \cdot \frac{1}{\sin^2 \alpha}$$

$$= \frac{2 \sin \alpha \cos \alpha}{\sin^2 \alpha}$$

$$= \frac{2 \cos \alpha}{\sin \alpha}$$

$$= 2 \cot \alpha$$

using $S_{\infty} = \frac{a}{1-r}$

(9)

$$5(b) \left(x^2 - \frac{2}{x}\right)^6 = \dots + {}^6C_r (x^2)^{6-r} \left(\frac{2}{x}\right)^r + \dots$$

$$= \dots + {}^6C_r (x^2)^{6-r} (x^{-r}) + \dots$$

$$= \dots + {}^6C_r 2^r x^{12-2r-r} x^{-r} + \dots$$

$$= \dots + {}^6C_r 2^r x^{12-3r} + \dots$$

The term is independent of x for x^0 i.e. $12-3r=0$
 $\therefore r=4$

$\therefore$ required term ${}^6C_4 \cdot 2^4$

$$= \frac{6!}{4!2!} \cdot 16$$

$$= \frac{6 \cdot 5 \cdot 4!}{4!2!} \cdot 16$$

$$= 240$$

HE3

$$(c) \sqrt{3} \cos \theta - \sin \theta = 1$$

$$\therefore \frac{\sqrt{3}}{2} \cos \theta - \frac{1}{2} \sin \theta = \frac{1}{2}$$

$$\sin \alpha \cos \theta - \cos \alpha \sin \theta = \frac{1}{2}$$

$$\sin(\alpha - \theta) = \frac{1}{2}$$

$$\frac{\pi}{3} - \theta = \sin^{-1} \frac{1}{2}$$

$$\therefore \frac{\pi}{3} - \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = \frac{\pi}{3} - \frac{\pi}{6}, \frac{\pi}{3} - \frac{5\pi}{6}$$

$$= \frac{\pi}{6}, -\frac{2\pi}{6}$$

$$= \frac{\pi}{6}, -\frac{\pi}{2}$$

$$= \frac{\pi}{6}, \frac{3\pi}{2}$$

HE7

(5)

$$\left. \begin{aligned} \frac{\sqrt{3}}{2} \cos \theta - \frac{1}{2} \sin \theta &= \frac{1}{2} \\ \sin \alpha \cos \theta - \cos \alpha \sin \theta &= \frac{1}{2} \end{aligned} \right\} \text{ for } \sin \alpha = \frac{\sqrt{3}}{2}, \cos \alpha = \frac{1}{2} \Rightarrow \alpha = \frac{\pi}{3}$$

$$\sin(\alpha - \theta) = \frac{1}{2}$$

$$\frac{\pi}{3} - \theta = \sin^{-1} \frac{1}{2}$$

$$\therefore \frac{\pi}{3} - \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = \frac{\pi}{3} - \frac{\pi}{6}, \frac{\pi}{3} - \frac{5\pi}{6}$$

$$= \frac{\pi}{6}, -\frac{2\pi}{6}$$

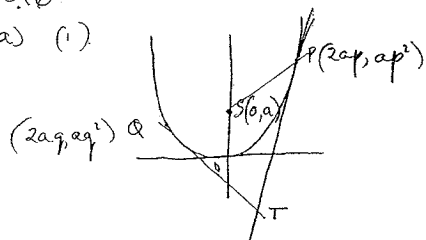
$$= \frac{\pi}{6}, -\frac{\pi}{2}$$

$$= \frac{\pi}{6}, \frac{3\pi}{2}$$

(4)

Q.6

(a) (i)



gradient PT :

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$

$$= \frac{2 \cdot 2ap}{4a}$$

$$= p$$

eq. PT :

$$y - ap^2 = p(x - 2ap)$$

$$y = px - 2ap^2 + ap^2$$

$$y = px - ap^2$$

(ii) eq. QT :

$$y = qx - aq^2$$

Pt. intersection :

$$y = px - ap^2$$

$$y = qx - aq^2$$

$$0 = (p - q)x - a(p^2 - q^2)$$

$$\Rightarrow (p - q)x = a(p^2 - q^2)$$

$$(p - q)x = a(p - q)(p + q)$$

$$x = a(p + q)$$

$$y = q[a(p + q)] - aq^2$$

$$= apq + aq^2 - aq^2$$

$$y = apq$$

co-ords of T : $(a(p + q), apq)$

(iii) SP² :

$$SP^2 = (2ap - 0)^2 + (ap^2 - a)^2$$

$$= 4a^2p^2 + a^2(p^2 - 1)^2$$

$$= 4a^2p^2 + a^2(p^4 - 2p^2 + 1)$$

$$= 4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2$$

$$= a^2(p^4 + 2p^2 + 1)$$

$$= a^2(p^2 + 1)^2$$

$$SP = a(p^2 + 1)$$

(iv) From (ii) SQ = a(q² + 1)

$$SP + SQ = a(p^2 + 1) + a(q^2 + 1)$$

$$= a(p^2 + q^2 + 2)$$

$$= 4a$$

$$\therefore p^2 + q^2 + 2 = 4$$

$$p^2 + q^2 - 2 = 0 \rightarrow p^2 + q^2 = 2$$

From (ii) $y = apq$ $x = a(p + q)$

$$pq = \frac{y}{a} \quad p + q = \frac{x}{a}$$

$$(p + q)^2 = \frac{x^2}{a^2}$$

$$p^2 + q^2 + 2pq = \frac{x^2}{a^2}$$

$$\therefore 2 + 2\left(\frac{y}{a}\right) = \frac{x^2}{a^2}$$

$$2a^2 + 2ay = x^2$$

$$x^2 = 4a\left(\frac{y}{2} + a\right) \text{ which is the equation}$$

of a parabola

(b). For SHM. $x = a \cos(nt + \alpha)$

$$x = 3 \cos\left(\frac{1}{2}t + \alpha\right)$$

Let θ have displacement $x = 0$

Initially at $t = 0, x = 0$

$$0 = 3 \cos(0 + \alpha)$$

$$\cos \alpha = 0 \therefore \alpha = \frac{\pi}{2}$$

$$x = 3 \cos\left(\frac{1}{2}t + \frac{\pi}{2}\right)$$

$$\dot{x} = -3 \sin\left(\frac{1}{2}t + \frac{\pi}{2}\right) \cdot \frac{1}{2}$$

$$= -\frac{3}{2} \sin\left(\frac{1}{2}t + \frac{\pi}{2}\right)$$

at $x = 0, \dot{x} = -\frac{3}{2} \sin\left(\frac{1}{2} \cdot 0 + \frac{\pi}{2}\right)$

$$= -\frac{3}{2} \sin \frac{\pi}{2}$$

$$= -\frac{3}{2} \text{ cm s}^{-1}$$

Speed is $\frac{3}{2} \text{ cm s}^{-1}$

$$T = \frac{2\pi}{n} \quad a = 3$$

$$4\pi = \frac{2\pi}{n}$$

$$\therefore n = \frac{1}{2}$$

Alternative

$$T = \frac{2\pi}{n} \therefore 4\pi = \frac{2\pi}{n} \rightarrow n = \frac{1}{2} \quad a = 3$$

$$v^2 = n^2(a^2 - x^2)$$

$$v^2 = \frac{1}{2}^2(3^2 - 0) \text{ at } x = 0$$

$$v^2 = \left(\frac{3}{2}\right)^2$$

$$v = \pm \frac{3}{2} \text{ m s}^{-1}$$

$\therefore$ speed = $\frac{3}{2} \text{ m s}^{-1}$ about $x = 0$

Q6(c)

$$(1) T = T_0 + C e^{kt} \rightarrow T - T_0 = C e^{kt}$$

$$\frac{dT}{dt} = k C e^{kt} = k(T - T_0)$$

$\therefore \frac{dT}{dt} \propto T - T_0$ satisfying the differential equation

$$(11) \text{ Initially } t=0, T_0 = 5, T = 20$$

$$20 = 5 + C e^{k(0)} = 5 + C$$

$$\therefore C = 15$$

$$\text{In general } T = 5 + 15 e^{kt}$$

$$\text{at } T = 17, t = 0.5 \text{ h}$$

$$17 = 5 + 15 e^{\frac{k}{2}}$$

$$e^{\frac{k}{2}} = \frac{12}{15} = 0.8$$

$$\frac{k}{2} = \ln 0.8$$

$$k = 2 \ln(0.8)$$

$$= -0.4463 \dots (4 \text{ dp})$$

HE3

$$\text{at } T = 10,$$

$$T = 5 + 15 e^{kt}$$

$$10 = 5 + 15 e^{-0.4463t}$$

$$\frac{5}{15} = e^{-0.4463t}$$

$$-0.4463t = \ln\left(\frac{1}{3}\right)$$

$$t = \frac{\ln\left(\frac{1}{3}\right)}{-0.4463}$$

$$= 2.4616$$

$$= 2.46 (2 \text{ dp})$$

4

Q7.

$$\begin{aligned} \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ &= \frac{d}{dx} (10x - x^2)^{1/2} \\ &= \frac{1}{2} (10x - x^2)^{-1/2} (10 - 2x) \\ &= \frac{(10 - 2x)}{2 \sqrt{10x - x^2}} \\ &= \frac{10 - 4}{2 \sqrt{20 - 4}} \quad \text{at } x = 2 \\ &= \frac{6}{2 \sqrt{16}} \\ &= \frac{3}{4} \\ &= 0.75 \text{ m s}^{-2} \end{aligned}$$

HE5

2

$$(1) {}^nC_r + {}^nC_{r+1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r+1)!(n-(r+1))!}$$

$$= \frac{n!(r+1)}{(r+1)r!(n-r)!} + \frac{n!}{(r+1)!(n-(r+1))!}$$

$$= \frac{n!(r+1)}{(r+1)!(n-r)!} + \frac{n!}{(r+1)!(n-(r+1))!}$$

$$= \frac{n!(r+1)}{(r+1)!(n-r)!} + \frac{n!(n-r)}{(r+1)!(n-r)!}$$

$$= \frac{n!}{(r+1)!(n-r)!} [r+1 + n-r]$$

$$= \frac{(n+1)n!}{(r+1)!(n-r)!}$$

$$= \frac{(n+1)!}{(r+1)!(n-(r+1))!}$$

$$= {}^{n+1}C_{r+1}$$

HE3

2

Q. 7(c)

(11). To prove $\sum_{i=3}^n i^{-1} C_2 = {}^n C_3$

ie ${}^2 C_2 + {}^3 C_2 + {}^4 C_2 + \dots + {}^{n-1} C_2 = {}^n C_3$ for $n \geq 3$

at $n=3$, LHS = ${}^{3-1} C_2 = {}^2 C_2 = 1$

RHS = ${}^3 C_3 = 1$.

$\therefore$ true for $n=3$.

(2)

Assume ${}^2 C_2 + {}^3 C_2 + {}^4 C_2 + \dots + {}^{n-1} C_2 = {}^n C_3$ is true for $n=k$, $k \geq 3$

$\therefore {}^2 C_2 + {}^3 C_2 + {}^4 C_2 + \dots + {}^{k-1} C_2 = {}^k C_3$

For $n=k+1$

${}^2 C_2 + {}^3 C_2 + {}^4 C_2 + \dots + {}^{k-1} C_2 + {}^k C_2 = {}^k C_3 + {}^k C_2$

$= {}^k C_2 + {}^k C_3$

$= {}^{k+1} C_3$ from (1)

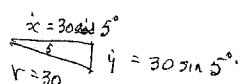
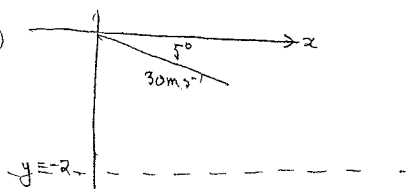
$\therefore {}^2 C_2 + {}^3 C_2 + \dots + {}^{k-1} C_2 = {}^k C_2$ is true for $n=k+1$

Since it is true for $n=3$, it is true for $n=3+1=4$, since it is true for $n=4$, it is true for $n=4+1=5$ and so on.

$\therefore \sum_{i=3}^n i^{-1} C_2 = {}^n C_3$ is true for all $n \geq 3$.

(b) (1)

HE3



(b)

Initially $\ddot{x} = 0$
 $\ddot{y} = c$

at $t=0$, $\dot{x} = 30 \cos 5^\circ = c$

$x = \int 30 \cos 5^\circ dt$
 $= 30t \cos 5^\circ$

(11) (cont)

$\ddot{y} = -10$

$\dot{y} = \int -10 dt$

$\dot{y} = -10t + c$

at $t=0$, $\dot{y} = 30 \sin 5^\circ \therefore 30 \sin 5^\circ = 0 + c \rightarrow c = 30 \sin 5^\circ$

$\dot{y} = -10t + 30 \sin 5^\circ$

$y = \int (-10t + 30 \sin 5^\circ) dt$

at $t=0$, $y=0 \therefore c_2 = 0$

$y = -5t^2 - 30t \sin 5^\circ$

(ii) The ball hits the ground at $y = -2$.

$\therefore -2 = -5t^2 - 30t \sin 5^\circ$

$5t^2 + (30 \sin 5^\circ)t - 2 = 0$

$t = \frac{-30 \sin 5^\circ \pm \sqrt{(30 \sin 5^\circ)^2 - 4 \cdot 5 \cdot (-2)}}{2(5)}$

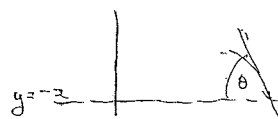
$= \frac{-30 \sin 5^\circ \pm \sqrt{(30 \sin 5^\circ)^2 + 40}}{10}$

$= \frac{-2.6146 \pm \sqrt{46.8858}}{10}$

$= \frac{-2.6146 + 6.8437}{10}$, $t > 0$.

$= 0.4229$
 $= 0.423 \text{ sec.}$

(iii)



at $t = 0.423 \text{ s}$, $\dot{x} = 30 \cos 5^\circ = 29.8858$

$\dot{y} = -10(0.423) + 30 \sin 5^\circ$
 $= -2.6146 + 2.6146$

$\tan \theta = \left| \frac{\dot{y}}{\dot{x}} \right| = \left| \frac{-2.6146}{29.8858} \right|$

$\theta = 13.23^\circ$

$\approx 13^\circ$ (nearest degree)